

Software Operation

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Introduction

This section presents detailed operating instructions for both the BioFocus 2000 and the BioFocus 3000 CE systems. The instructions cover general operating procedures that are common to all methodologies.

Before operating your BioFocus CE system, it is important that you are familiar with the safety features described earlier, and that the system has been installed in accordance with the procedures described.

You must have a working knowledge of Microsoft Windows 3.1 or higher to operate the BioFocus CE systems. If you are not already familiar with Windows, please refer to the Windows User's Guide for details.

Tips on Programming Automation Sequences

An automation sequence provides the ability to combine several methods in a single program, have random access to all vial positions. Also, you have the ability to save configurations, methods, and automation sequences for future reference and use.

To simplify programming of an automation sequence for first time users, the program database supplies predefined Configuration, Method and Autosequence templates for running different Bio-Rad chemistries (CZE, IEF, PCR, and SDS-Protein).

Programming an automation sequence through the Main Screen buttons should be done in the following order:

Reagents – to add new reagents to the data base.

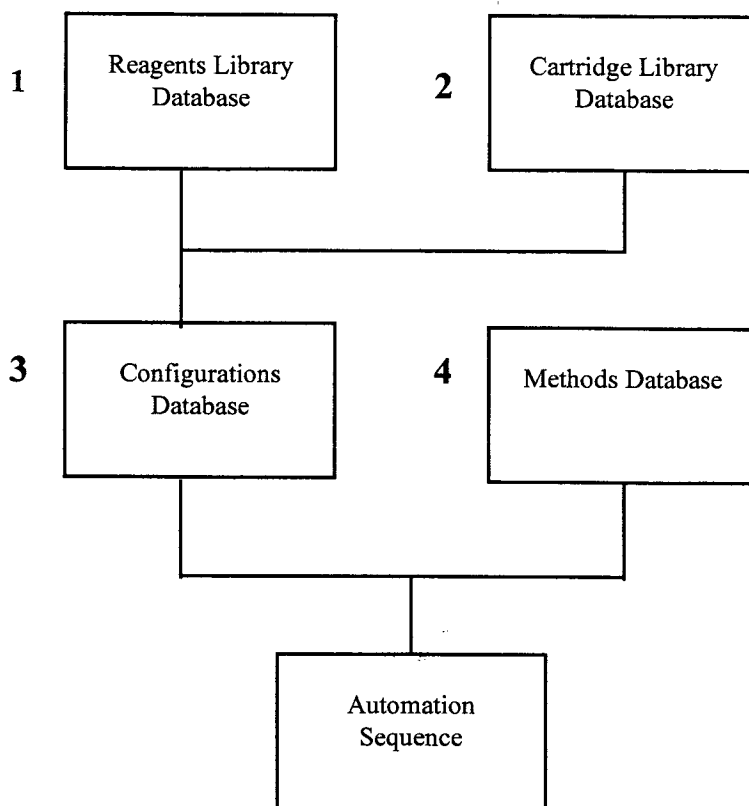
Cartridges – to add new Cartridges to the data base.

Configurations – to define a new Configuration using one of the configuration templates in the data base.

Methods – to define a new Method using one of the method templates in the data base.

Auto Seqs – to program a new Automation Sequence that links new Methods to the new Configuration using one of the autosequence templates in the data base.

The next sections step through the various screens necessary to define, schedule and execute an automation sequence for performing basic CZE. You will use the reference Configuration, Method, and Autosequence templates supplied in the data base. Refer to the diagram below, which gives an overview of the sequence of operations.



Turning on the System

Absorbance Mode

NOTE: If you have installed the LIF² Detector and connected the BioFocus COM1 cable to the LIF² Detector, you must also turn on power to the LIF² Detector.

- Turn on the computer's power switch.
- Turn on the power switch on the rear panel of your BioFocus instrument. Both doors should be closed. The BioFocus will now ensure that the vial holders are lowered and that the carousels are in their home (number 1) positions.

Establishing Communication

If the DOS command line prompt appears, type WIN and press the **Enter** key. Next, double-click the Bio-Rad BioFocus icon in the main Windows screen, then click the BioFocus icon.

If your computer is running Windows 95, the main menu comes up automatically after you turn on power to the computer. Click the **Start** button, select **Programs**, select Bio-Rad BioFocus, and then click the BioFocus icon.

Note: BioFocus 2000 detector power-up requires approximately 75 seconds.

An hourglass appears while the computer is checking detector status and opening communications with the BioFocus. If the BioFocus detector is not on, or is not connected properly, a warning message appears. You must use the mouse to select an option: click **Abort**, **Retry**, or **Ignore**. See Figure 5-1.

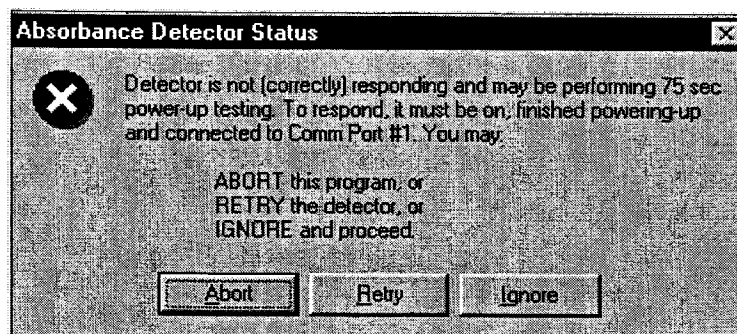


Figure 5-1 BioFocus Detector Status Dialog Box

If the BioFocus system is not on, or is not properly connected, or there is a subsystem problem, a warning message appears. You must use the mouse to select an option: click **Abort**, **Retry**, or **Ignore**. See Figure 5-2.

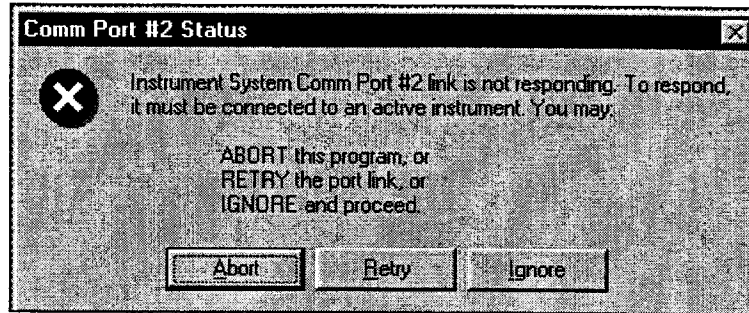


Figure 5-2 BioFocus Not Responding Dialog Box

If your BioFocus system is equipped with carousel temperature control (BioFocus 2000TC or BioFocus 3000TC), the Carousel Temperature Control screen appears after the computer verifies that both the detector and BioFocus system are ready for operation,. See Figure 5-3.

NOTE: You may access this panel using the carousel temperature button to change the carousel temperature after the completion or termination of an automation sequence, but not while one is underway.



Carousel Temperature Button

NOTE: If the ambient temperature exceeds 30°C, the carousel temperature cannot cool to 4° C. At 40° C ambient temperature, the carousel compartment cannot cool to lower than 15° C, regardless of the setpoint.

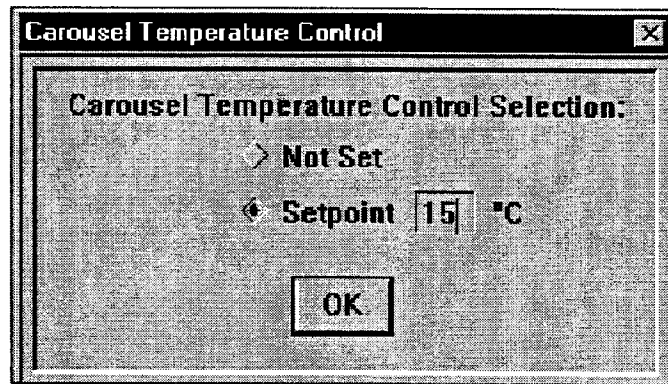


Figure 5-3 Carousel Temperature Control Dialog Box

Allow approximately one-half hour for the carousel compartment to reach the temperature you selected before starting an automation sequence. A half hour waiting period also allows time for the detector to stabilize. If you are planning to cool labile samples, we also recommend that you store your carousels in a refrigerator when not in use. Install them in the BioFocus system just before starting an automation sequence.

At start-up, after entering your carousel temperature selection, the BioFocus control program automatically initiates a Database Cleanup to verify database integrity and to permanently remove all data base files that were previously deleted. The program then proceeds to the BioFocus Main Screen shown in Figure 5-6.

LIF Mode

System startup in LIF mode is very similar to startup in absorbance mode. The software detects that you moved the absorbance detector from its operational position into the storage position. Only those screens and commands that pertain to LIF operation will be available.

- Turn on the computer's power switch.
- Turn on the power switch on the rear panel of your BioFocus instrument. Both doors should be closed. If you turned off the system earlier while vial holders were in the upper position or before carousels had completed their rotations, carousels will now return to "home" position.
- Turn on the LIF² Detector power switch.

Establishing Communications

If the DOS command line prompt appears, type WIN and press the **Enter** key. Next, double-click the Bio-Rad BioFocus icon in the main Windows screen, then click the BioFocus icon.

If your computer is running Windows 95, the main menu comes up automatically after you turn on power to the computer. Click the **Start** button, select **Programs**, select Bio-Rad BioFocus, and then click the BioFocus icon.

An hourglass appears while the computer is checking detector status and opening communications with the BioFocus. If the BioFocus detector is not on, or is not connected properly, a warning message appears. You must use the mouse to select an option: click **Abort**, **Retry**, or **Ignore**. See Figure 5-1.

If the computer cannot establish communications with the BioFocus instrument or the LIF² Detector, a similar series of messages appears as those described for absorbance mode. See Figure 5-2.

After the computer verifies that both the LIF² Detector and BioFocus are ready for operation, the LIF² Detector temperature and laser control

screen appears. See Figure 5-4. Enter carousel temperature settings as for absorbance mode. The correct laser configuration should be identified automatically, however.

You must enter values for expected wavelengths of laser-induced emissions from native or tagged sample components. If you do not enter values, an error message appears. You cannot move to the next screen without entering emission wavelength values.

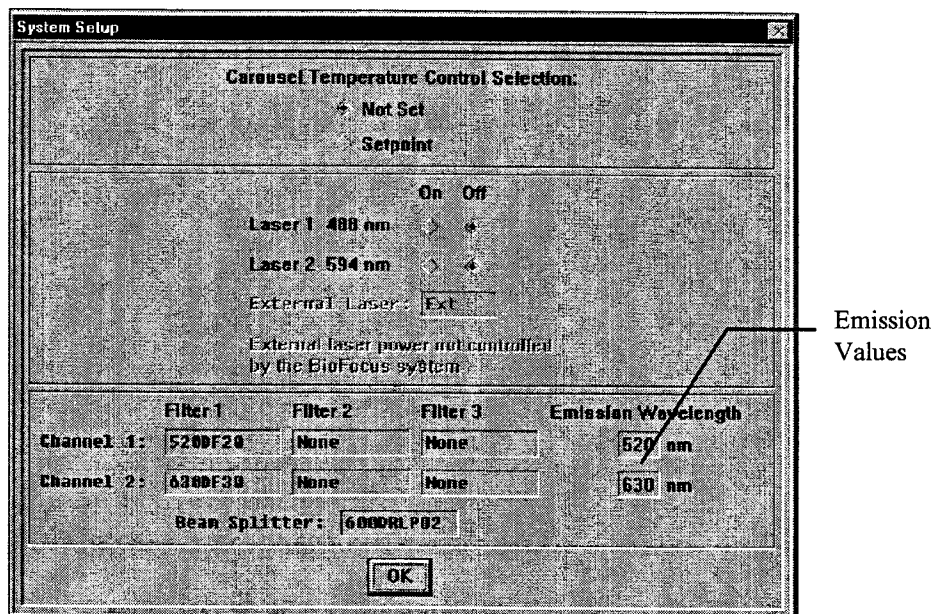


Figure 5-4 Laser and Temperature Set Screen

Allow approximately one-half hour for the carousel compartment to reach the temperature you selected before starting an automation sequence. A half hour waiting period also allows time for the detector to stabilize. If you are planning to cool labile samples, we also recommend that you store your carousels in a refrigerator when not in use. Install them in the BioFocus system just before starting an automation sequence.

At start-up, after entering you carousel temperature selection, the BioFocus control program automatically initiates a Database Cleanup to verify database integrity and to permanently remove all data base files that were previously deleted. The program then proceeds to the BioFocus Main Screen shown in Figure 5-6.

Simulation Mode

If the software cannot establish communications with the BioFocus instrument or cannot check detector status, you can operate in Simulation mode. Click **Ignore** in both screens that appear when the software fails to communicate or check status. A screen then appears to give you a

choice of simulation temperature control zones and detectors. See Figure 5-5. Click the choice that you wish, either:

- Cartridge and Carousels, or Cartridge only
- Absorbance or LIF² detector

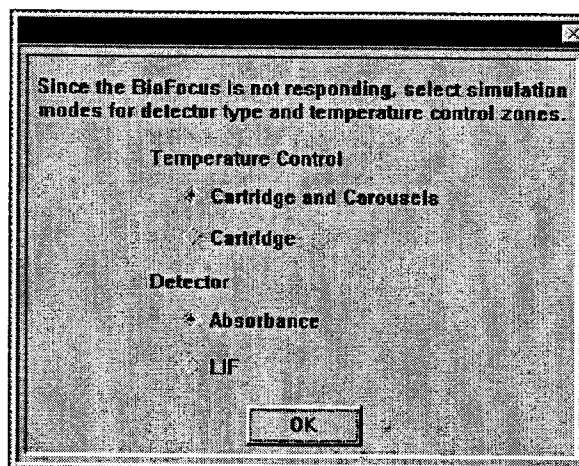


Figure 5-5 Simulation Mode Screen

Make your selection and click **OK** to bring up the next screen, which allows you to choose a temperature setpoint screen and laser/filter settings if running the simulated LIF mode. Click **OK** to bring up screens that notify you of data base cleanup. After cleanup, the main BioFocus screen appears. You may then perform method and automation sequence definition and simulate operation. Autosequences may be simulated only if the detector is operational.

Demonstration Mode

This mode is similar to Simulation mode, but you may enter Demonstration mode without a BioFocus instrument connected to the computer. To use Demonstration mode, use this procedure:

For Windows 3.1x:

1. Pull down the **File** menu and select **Run**.
2. Type `C:\CE2000\CE2000.exe dmo` or `C:\CE3000\CE3000.exe dmo` and press the **Enter** key. This starts the BioFocus software in Demonstration mode.

For Windows 95:

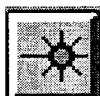
1. Click **Start**.
2. Click **Run**.
3. Type `C:\CE2000\CE2000.exe dmo` or `C:\CE3000\CE3000.exe dmo`.
4. Click **OK**. This starts the BioFocus software in Demonstration mode.

This brings up a screen that allows you to choose temperature control and either absorbance or LIF mode. Click your choice and click **OK**. This brings up the main screen. Depending on your choice of operating mode, only those screens appear that apply to the mode chosen.

BioFocus Main Screen



UV/Vis Diagnostic Button



LIF Diagnostic Button

The Main Screen consists of a Menu Bar, a Toolbar, the Main Screen buttons, and a Status Bar. The screen shown in Figure 5-6 appears in UV/Vis mode. In LIF mode, the screen shows a laser symbol in the light level diagnostics button instead of an incandescent light symbol.

Functions found under the Main Screen Menu Bar are listed below:

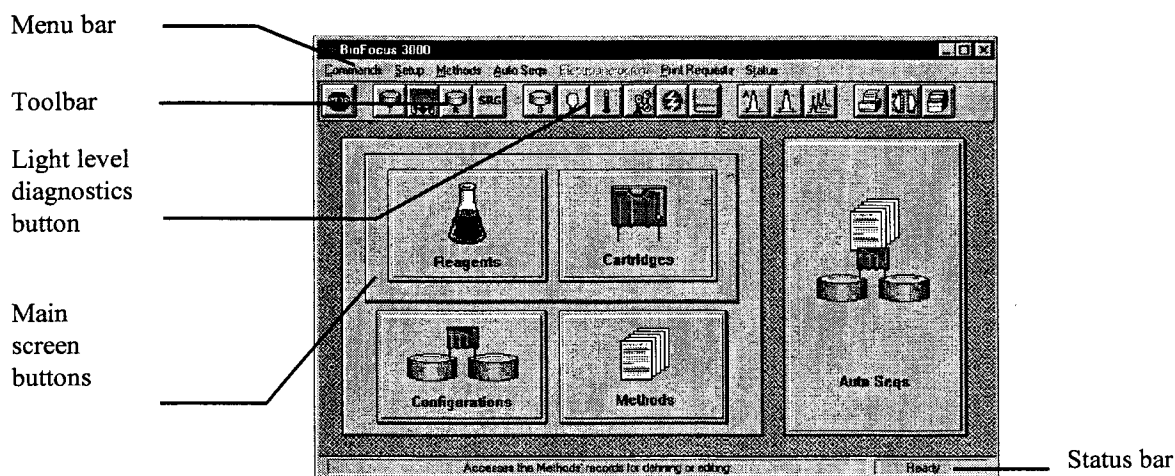


Figure 5-6 BioFocus Main Screen

Menu Bar Functions

Commands – You can begin or end a scheduled automation sequence, reset communications between the BioFocus instrument and the computer, monitor the baseline, perform diagnostic tests, autocalibrate the detector (BioFocus 3000 absorbance mode only), reset carousels, and change the carousel temperature in between automation sequences. You can access Single Run Group Methods, request or terminate integration during autosequence, access post-run integration, clean up the database (hard disk), keep the vials up at the end of an automation sequence, get information about the BioFocus, or exit the BioFocus program from this menu. Most of these commands can also be accessed through the Toolbar buttons.

NOTE: All functional menu commands have darkened letters. When a menu command is “grayed out” (lightened), it is nonfunctional in the current situation

Setup – You can add, edit or delete reagents and cartridges in the database and define, edit or delete configurations from this menu.

Methods – You can define, edit or delete methods. You can select all operating parameters: voltage, current, polarity, type and time of injection, run buffers, capillary purges, detection wavelength(s) and sensitivity, run time, and cartridge temperature.

Auto Seqs – You can link a configuration with one or more methods to form run groups. You can define a single sample or run group, or you can link to multiple methods and run groups before scheduling or starting an automation sequence.

Electropherogram – This command brings the electropherogram of any on-going run back into view after checking status screens or multitasking in other windows.

Print Requests – You can make hard copies of the Cartridge and Reagent Lists and any BioFocus Method, Automation Sequence or Configuration in the database library.

Status – You can use this command to check the status of BioFocus subsystems.

The commands on the main screen menu bar and under each item have one character underscored. For the menu bar itself, you may hold down the **Alt** key and type this character on the keyboard to call up a screen as an alternative to clicking the command with the mouse. For commands under each menu bar item, type the underscored character without holding down the **Alt** key.

Toolbar Functions

You can also access most Menu Bar commands through the Main Screen Toolbar. In addition to these commands, the Main Screen Toolbar also contains several functions which are not available under **Commands**. These are used to access Spectra (for the BioFocus 3000) or View Electropherogram (for the BioFocus 2000).



Spectra Button



*View Electropherogram
Button*

The only controls listed under **Commands** which are not accessible through the toolbar buttons are: Database Cleanup, Detector Autocalibrate (BioFocus 3000 only), Exit from the BioFocus program, and About BioFocus. The main screen toolbar for a BioFocus 3000 system with carousel temperature control and absorbance capability is shown in Figure 5-7.

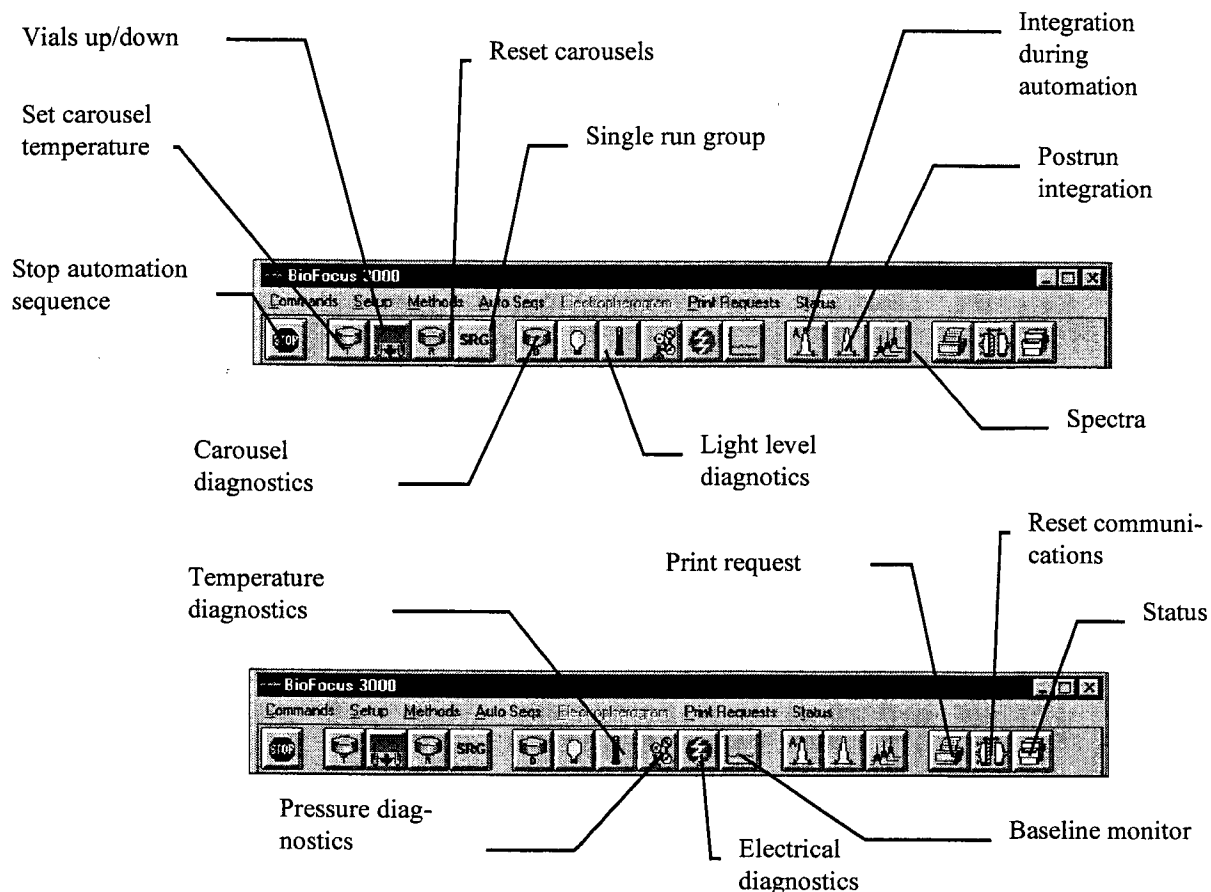


Figure 5-7 BioFocus 3000 Toolbar



Example of a Button Label

Toolbar items can be accessed by a simple click on the toolbar buttons. For ease of use, each toolbar button is also equipped with a Button Label that describes the button's function. The button label appears when the cursor is moved over the desired button on the toolbar. When a toolbar button is inactive, this means that it is nonfunctional in the current situation. If the toolbar button is active, the cursor becomes a hand; if it is inactive, then the cursor becomes an arrow.

Main Screen Buttons

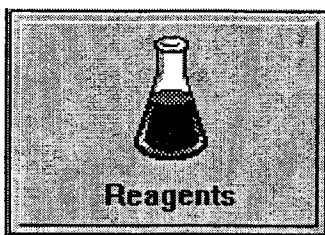
Main screen buttons (and some menus) allow you to edit or define reagents, cartridges, configurations, methods, and automation sequences.

Status Bar

When you place the mouse pointer on any toolbar or main screen button, the Status Bar at the bottom left corner of the main screen displays the description for that particular item. The Status bar at the bottom right displays current system status (ready, executing, priority, etc.).

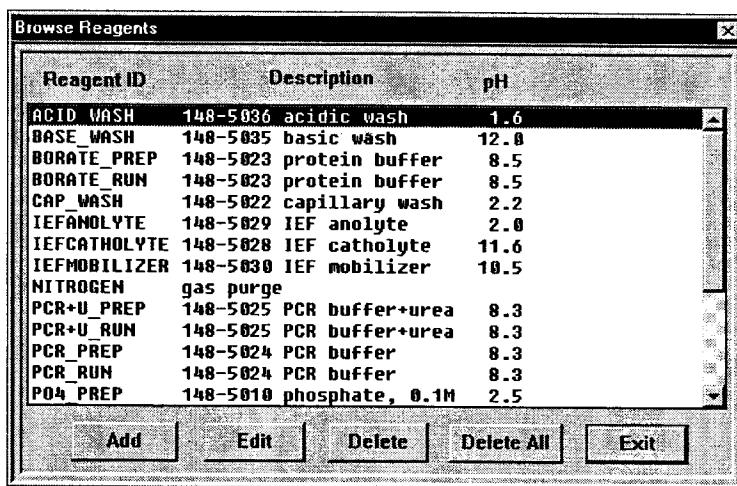
Detailed Programming Instructions

Modifying the Reagent Data Base



Reagents Button

Click the **Reagents** button (you can also access Reagents from the **Setup** menu on the Menu Bar) and you will see a screen (Figure 5-8) where you can add, edit or delete reagents in the database. Each buffer or wash solution you plan to use and each internal standard you plan to inject by Multi Inject (see “Multi Inject” on page 39) must have a unique ID and be listed in this Browse Reagents screen.



Reagent ID	Description	pH
ACID WASH	148-5036 acidic wash	1.6
BASE WASH	148-5035 basic wash	12.0
BORATE_PREP	148-5023 protein buffer	8.5
BORATE_RUN	148-5023 protein buffer	8.5
CAP WASH	148-5022 capillary wash	2.2
IEFANOLYTE	148-5029 IEF anolyte	2.0
IEFCATHOLYTE	148-5028 IEF catholyte	11.6
IEFMobilizer	148-5030 IEF mobilizer	10.5
NITROGEN	gas purge	
PCR+U_PREP	148-5025 PCR buffer+urea	8.3
PCR+U_RUN	148-5025 PCR buffer+urea	8.3
PCR_PREP	148-5024 PCR buffer	8.3
PCR_RUN	148-5024 PCR buffer	8.3
P04 PREP	148-5010 phosphate, 0.1M	2.5

Figure 5-8 Browse Reagents Dialog Box

Predefined Reagents

In the database provided with the instrument, there is a list of CE reagents available from Bio-Rad with their reagent Ids. These reagent listings can be used as templates. The information in the rest of the boxes can be left the same as the template or changed as needed. The description and pH values are for information only and serve as a reference for the BioFocus system user.

Multiple vials of identical solution for frequent buffer changes or sequential washes must be uniquely labeled, such as Buffer_A, Buffer_B, Buffer_C, etc.

Adding a New Reagent

To add a new reagent to the database, click the reagent that you want to use as a template, then click the **Add** button at the bottom of the screen. You must assign a unique ID (in Figure 5-9 it's "ACID_WASH_A") and click **OK**.

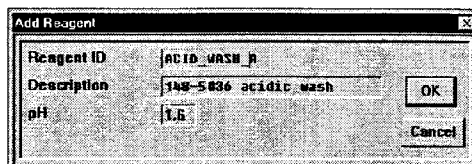


Figure 5-9 Add Reagent Dialog Box

If you forget to assign a unique name, a message appears requesting one. See Figure 5-10.

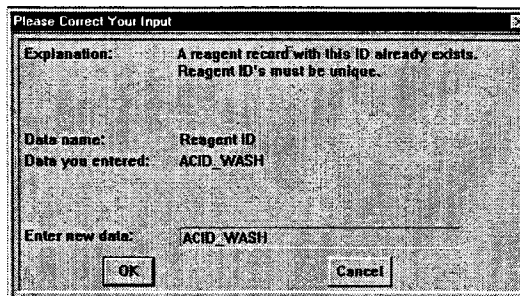


Figure 5-10 Correct Input Dialog Box

Editing a Database Entry

After you have created a listing, you may edit it by clicking **Edit**. Figure 5-11 shows a reagent browser that can be used for editing.

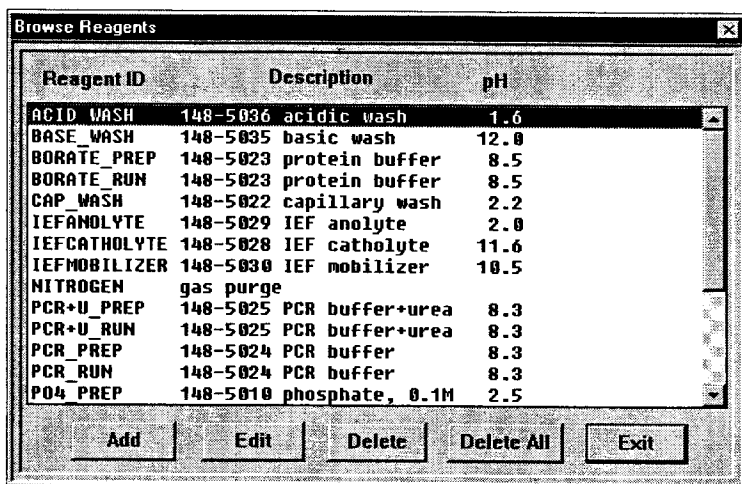
NOTE: A reagent may not be edited, deleted or autodeleted if it is already linked to any other record(s) in the database (configuration, method or automation sequence).

Non-Editable Reagent Templates

In the database provided with the instrument, Bio-Rad supplies two different types of reference reagent templates: non-editable and editable. The IEFANOLYTE, IEF CATHOLYTE, IEF MOBILIZER, CAP_WASH, NITROGEN, PO4_PREP, PO4_RUN and WATER reference reagent templates are noneditable and nondeletable from the database.

Editable Reagent Templates

Other reagent templates in the database such as BORATE, PCR, and SDS-PROTEIN, are not universally noneditable and nonautodeletable (cannot be deleted with the **Delete All** button), but are singly-deletable unless they are linked to one or more record(s) in the database.



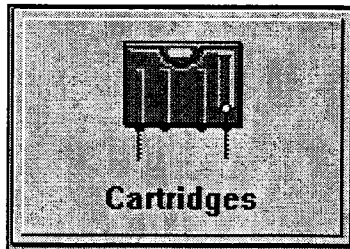
Reagent ID	Description	pH
ACID_WASH	148-5036 acidic wash	1.6
BASE_WASH	148-5035 basic wash	12.0
BORATE_PREP	148-5023 protein buffer	8.5
BORATE_RUN	148-5023 protein buffer	8.5
CAP_WASH	148-5022 capillary wash	2.2
IEFANOLYTE	148-5029 IEF anolyte	2.0
IEFCATHOLYTE	148-5028 IEF catholyte	11.6
IEFMOBILIZER	148-5030 IEF mobilizer	10.5
NITROGEN	gas purge	
PCR+U_PREP	148-5025 PCR buffer+urea	8.3
PCR+U_RUN	148-5025 PCR buffer+urea	8.3
PCR_PREP	148-5024 PCR buffer	8.3
PCR_RUN	148-5024 PCR buffer	8.3
PO4_PREP	148-5010 phosphate, 0.1M	2.5

Buttons: Add, Edit, Delete, Delete All, Exit

Figure 5-11 Browse Reagents Dialog Box

Click **Exit** to leave this screen.

Modifying the Cartridges Database



Cartridges Button

Overview

To view all cartridges in the database, click the **Cartridges** button (you can also access Cartridges from the **Setup** menu on the Menu Bar). User-Assembled Cartridges recommended for IEF, SDS-Protein, oligonucleotide, 1000 bp DNA, 4000 bp DNA applications and system verification are included as templates for your convenience. Figure 5-12 shows a Browse Cartridges screen.

Catalog Number	Serial Number	Length (cm)	Diameter (um)	Coated (Y/N)	Use Count	Active (Y/N)
148-3030	STD030	17	25	Y	0	N
148-3031	PEPTID	24	25	Y	0	Y
148-3031	STD031	24	25	Y	0	N
148-3032	STD032	36	50	Y	0	N
148-3033	STD033	50	50	Y	0	N
148-3035	STD035	24	50	Y	0	N
148-3040	STD040	50	50	N	0	N
UAC	IEF	24	50	Y	0	Y
UAC	PCR	24	75	Y	0	Y
UAC	SDS	24	50	N	0	Y
UAC	VERIFY	24	50	N	0	Y

Buttons: Add, Edit, Delete, Delete All, Act/Deact, Exit

Figure 5-12 Browse Cartridges Dialog Box

Adding a Cartridge

To add a cartridge, click the cartridge that you want to use as a template, then click the **Add** button at the bottom of the screen. In the example shown in Figure 5-13, the standard template for a Bio-Rad Catalog #148-3030 cartridge has been selected.

Add Cartridge

Standard Cartridge ☒ NON-STD ☐

Catalog Number: 148-3030 Serial Number: STD030

Length: 17 cm Diameter: 25 um ☒ Coated

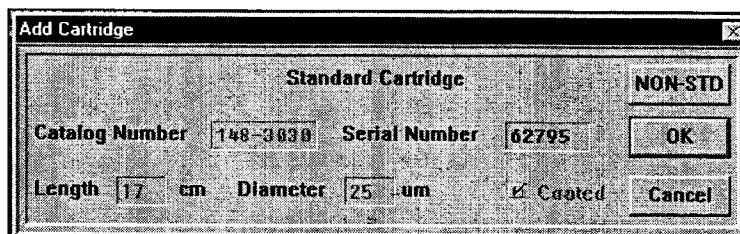
Buttons: OK, Cancel

Figure 5-13 Add Cartridge Dialog Box

Cartridge Serial Numbers and Use Count

You can track each cartridge that you have used simply by entering its serial number in the database with the appropriate template. Each cartridge must have a unique serial number. You may enter a name or any set of numbers (one to six characters). Refer to Figure 5-14.

The software automatically counts the number of times a cartridge has been used in an automation sequence and associates this with the serial number. This information may be useful in determining the suitability of a particular cartridge for further runs.

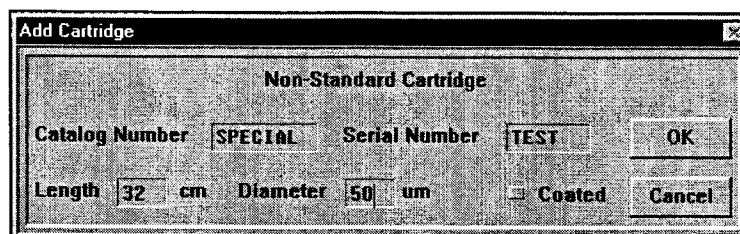


The image shows a dialog box titled "Add Cartridge". It has a tabbed interface with "Standard Cartridge" selected. The "NON-STD" tab is also visible. The "Catalog Number" field contains "148-3838" and the "Serial Number" field contains "62795". The "Length" field is "17" cm and the "Diameter" field is "25" um. There is a checked checkbox for "Coated". "OK" and "Cancel" buttons are at the bottom right.

Figure 5-14 Add Cartridge Dialog Box, Serial Number Entry

Adding a User-Assembled Cartridge

If you wish to add a User-Assembled Cartridge to the database, clicking the **NON-STD** button makes all of the information boxes available for editing (or you can simply use one of the application-specific User-Assembled Cartridges in the database as a template). See Figure 5-15.



The image shows a dialog box titled "Add Cartridge". It has a tabbed interface with "Non-Standard Cartridge" selected. The "Catalog Number" field contains "SPECIAL" and the "Serial Number" field contains "TEST". The "Length" field is "32" cm and the "Diameter" field is "50" um. There is an unchecked checkbox for "Coated". "OK" and "Cancel" buttons are at the bottom right.

Figure 5-15 Add Cartridge Dialog Box, Non-Standard Capillary Entry

Editing a Cartridge Entry

You may also edit cartridge information by selecting the cartridge to be modified and clicking the **Edit** button at the bottom of the Browse Cartridge screen shown in Figure 5-16, unless that cartridge is linked to a configuration.

Catalog Number	Serial Number	Length (cm)	Diameter (um)	Coated (Y/N)	Use Count	Active (Y/N)
148-3030	STD030	17	25	Y	0	N
148-3031	PEPTID	24	25	Y	0	Y
148-3031	STD031	24	25	Y	0	N
148-3032	STD032	36	50	Y	0	N
148-3033	STD033	50	50	Y	0	N
148-3035	STD035	24	50	Y	0	N
148-3040	STD040	50	50	N	0	N
UAC	IEF	24	50	Y	0	Y
UAC	PCR	24	75	Y	0	Y
UAC	SDS	24	50	N	0	Y
UAC	VERIFY	24	50	N	0	Y

Figure 5-16 Browse Cartridges Dialog Box

Deleting a Cartridge Entry

Whenever you wish to delete a cartridge from a database library, simply click it to highlight it, then click the **Delete** button. Once a cartridge has been linked to a configuration, it cannot be edited or deleted unless all other linked records (old and new automation sequences) are deleted first.

In the database provided with the instrument, Bio-Rad supplies two types of reference cartridge templates. The STD030, STD031, STD032, STD033, STD035 and STD040 standard reference cartridge templates are noneditable, nondeletable and nonautodeletable from the database and cannot be used in a configuration. The PEPTID, IEF, PCR, SDS, and VERIFY cartridge templates are also noneditable and nonautodeletable, but are singly-deletable unless they are linked to one or more records in the database and can be used in a configuration.

Deleting All Cartridge Entries

Clicking the **Delete All** button deletes all listed items except those items that are linked or are Bio-Rad supplied reference templates.

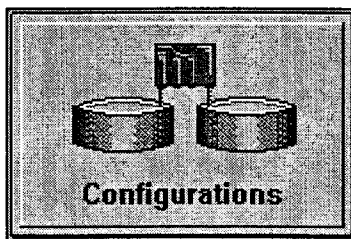
Activating or Deactivating a Cartridge

A cartridge can be made available or unavailable for a new configuration by either activating or deactivating it. This is done by highlighting the desired cartridge and then clicking the **Act/Deact** button. “STD” cartridge templates cannot be activated. You must have at least one cartridge activated before proceeding to Configurations.

Click **Exit** to leave this screen.

Modifying Configurations

A Configuration maps the physical setup of the instrument: what is in each carousel vial position (sample, reagent, internal standard, waste or fraction), and which capillary cartridge is being used.



Configurations Button

Viewing All Configurations

Click the **Configurations** button (you can also access Configurations from **Setup** on the Main Screen Menu Bar) to show all the configurations in the database. Figure 5-17 shows an example of a configuration list. The first time you use the BioFocus program, this screen displays a list of configuration templates for performing CZE, IEF, PCR, and SDS-Protein analysis. The screen also displays the System Verification template, which the software uses in running a system verification autosequence.

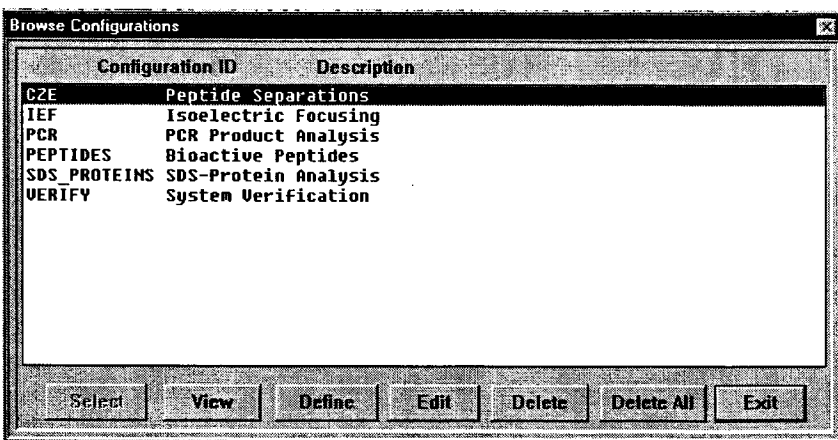


Figure 5-17 Browse Configurations Dialog Box

Each configuration template supplied with the instrument is editable and deletable only after deletion of all automation sequences to which it is linked.

Viewing Configuration Details

To view the details of an existing configuration, first highlight the configuration you wish to view and then click the **View** button. Clicking the **View** button brings up the View Configuration screen (see Figure 5-18) with its template parameters in place.

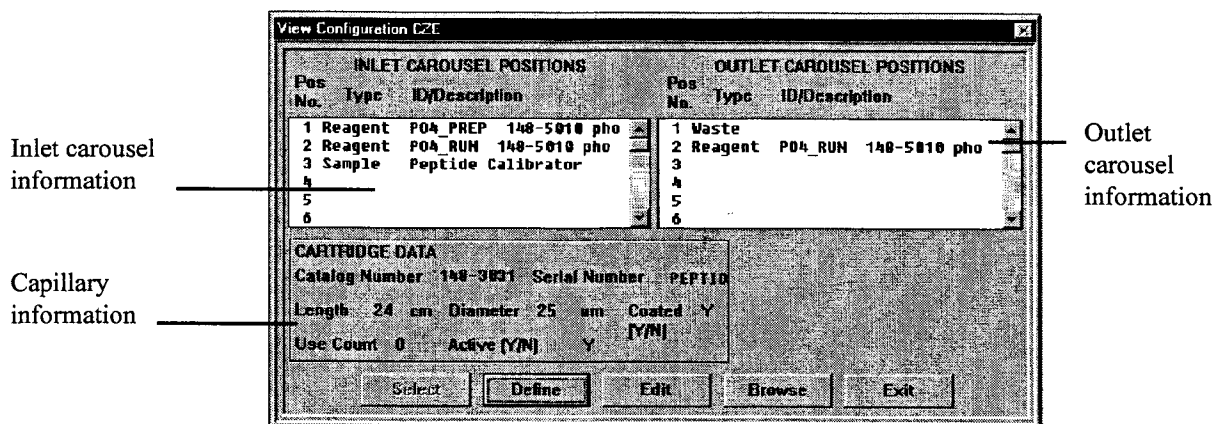


Figure 5-18 View Configuration CZE Screen

The screen above shows the View Configuration screen for the reference CZE template.

Defining a New Configuration

To define (add) a new template to the data base, click an existing template in the Browse Configurations screen to highlight it and then click **Define**. Careful selection of the configuration used as a template minimizes the amount of data entry necessary to define a new configuration. See Figure 5-19.

Before you begin to define a configuration, make sure that all reagents and the cartridge you will need have been added to the database.

Configuration ID and Cartridge (Screen 1)

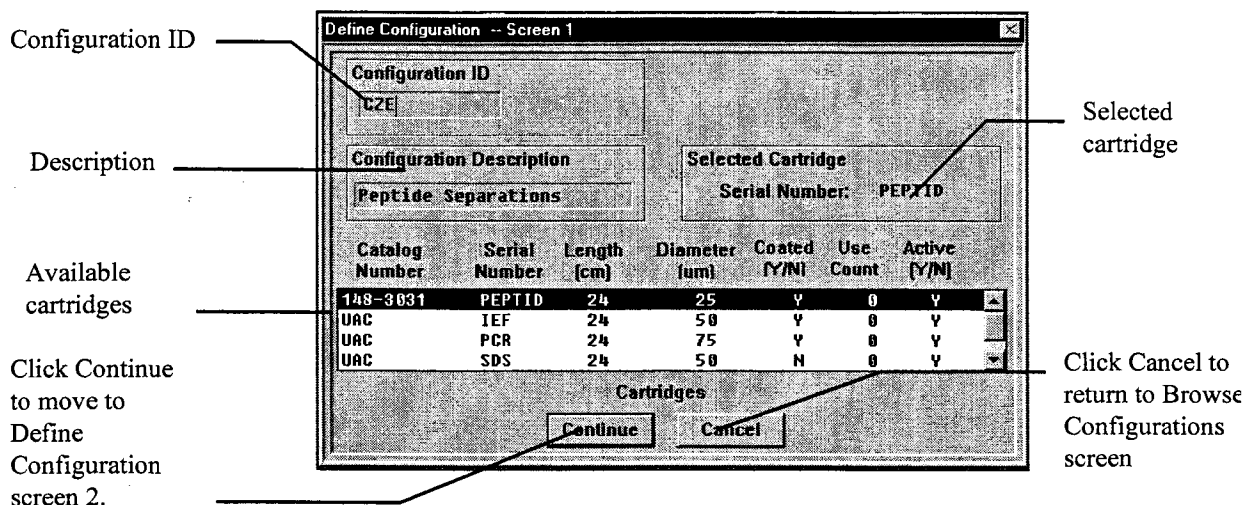


Figure 5-19 Define Configuration Screen 1

To define a configuration, first enter the Configuration ID and cartridge details. Use the following procedure:

1. Enter a unique configuration ID. It can be up to 12 characters long, without blank spaces.
2. Enter a configuration description if desired. Although not required, a description may help you differentiate between configurations with similar IDs. A description can be up to 25 characters long.
3. Select the cartridge for the Configuration from the list box on the screen. This box shows only cartridges in the database that are active. The example above shows four active cartridges in the database. Select one by double-clicking the appropriate cartridge. This cartridge should now appear in the Selected Cartridge box. If the desired cartridge is not present in the list box, you should cancel this Configuration definition, return to the Browse Cartridge screen, and either activate or add the desired cartridge.
4. Click **Continue** to move to Screen 2 and define the inlet carousel configuration.

Define Configuration -- Screen 1

Configuration ID
PEPTIDES

Configuration Description
Bioactive Peptides

Selected Cartridge
Serial Number: PEPTID

Catalog Number	Serial Number	Length (cm)	Diameter (um)	Coated (Y/N)	Use Count	Active (Y/N)
148-3031	PEPTID	24	25	Y	0	Y
UAC	IEF	24	50	Y	0	Y
UAC	PCR	24	75	Y	0	Y
UAC	SDS	24	50	N	0	Y

Cartridges

Continue Cancel

Figure 5-20 Define Configuration Screen 1, With Edited Configuration Description

Inlet Carousel (Screen 2)

→Tip:

Duplicate entries of the same Reagent ID are not allowed in the outlet carousel (a message appears to notify you that the reagent is already in the carousel). If you wish to use multiple vials for the same reagent in the outlet carousel, be sure to assign a unique name to each vial. For example, multiple phosphate buffer vials in the outlet carousel might be given a designation such as PO4_RUN, PO4_RUN_A, or PO4_RUN_B.

Define Configuration Screen 2, shown in Figure 5-21, allows you to designate what each vial contains and its position in the inlet carousel. Any of the preset parameters in Screen 2 can be changed when defining a new configuration.

Define Configuration PEPTIDES1 -- Screen 2

Define Inlet Carousel

Reagent

Sample

Empty

Reagent ID Description pH

ACID WASH	148-5036 acidic w	1.6
BASE WASH	148-5035 basic wa	12.0
BORATE PREP	148-5023 protein	8.5
BORATE RUN	148-5023 protein	8.5

Pos No. Type ID/Description

1	Reagent	PO4 PREP	148-5010 pho
2	Reagent	PO4 RUN	148-5010 pho
3	Sample	Peptide Calibrator	
4			
5			
6			
7			
8			

Inlet Carousel Positions

Continue Cancel Clear

Reagents

Available reagents

Vial positions

Figure 5-21 Define Configuration Screen 2 Inlet Carousel

→Tip:

If you plan to use Multi Inject, you must assign the internal standards you plan to inject to vial positions in the inlet carousel.

→Tip:

You may wish to assign deionized water to several reagent positions for washing the capillary and electrode in between capillary purging and sample injection. This is strongly recommended for accurate quantitation.

Assigning a New Reagent To a Vial Position

To assign a new reagent to a vial position you must first click the **Reagent** button at the upper left corner of the screen. Next use the mouse to double-click a reagent from the database list box in the lower left corner. That reagent now appears to the right of the **Reagent** button. Now double-click the Inlet Carousel Position (right side of screen) where you will place the reagent vial. The selected reagent now appears in that position.

Assigning a Sample To a Vial Position

To assign a sample to a vial position, click the **Sample** button, type in the sample name or description, then move the mouse to the right of the screen and double-click the carousel position where you wish to place the sample vial. Up to 31 sample vials may be assigned. In general, buffers may be placed anywhere in the inlet carousel, but all samples in any given run group must be in a set of contiguous vial positions. A configuration must include at least one Reagent and one Sample position in the inlet carousel.

Because samples will be linked to methods later in the program, you may have duplicate entries of any sample ID placed anywhere in the inlet carousel. Refer to Figure 5-23.

Deleting a Vial Assignment

To delete an existing vial assignment, first click **Empty** and then double-click the desired carousel vial position. Clicking the **Clear** button deletes all vial assignments in the inlet carousel.

Editing an Existing Configuration

If you create a template and then wish to edit it later, click **Edit**. This brings up the screen shown in Figure 5-22. Make any changes required and click **Continue** to access the next screen.

Edit Configuration IEF2 -- Screen 1

Configuration ID: IEF2

Configuration Description: Isoelectric Focusing

Selected Cartridge: Serial Number: IEF

Catalog Number	Serial Number	Length (cm)	Diameter (mm)	Coated (Y/N)	Use Count	Active (Y/N)
148-3831	PEPTID	24	25	Y	0	Y
UAC	IEF	24	50	Y	0	Y
UAC	PCR	24	75	Y	0	Y
UAC	SDS	24	50	N	0	Y

Cartridges

Continue **Cancel**

Figure 5-22 Edit Configuration Screen

Deleting a Configuration

If you wish to delete a configuration from the data base, highlight the configuration by clicking it from the Browse Configuration screen. Click **Delete** to remove the configuration list.

NOTE: You cannot delete reference configurations or those that are linked to an automation sequence.

Delete All

To delete all configurations (except those that are nondeletable), click **Delete All** from the Browse Configuration screen. This brings up a dialog box that informs you that you are about to delete all configurations that are not linked or are reference templates. To exit without deleting configurations, click **Cancel**.

Assigning Names to Multiple Vials

It is possible (but not recommended) to use a single reagent vial for both prepping the capillary and as an inlet run buffer. If, however, you wish to use multiple vials for the same reagent in the inlet carousel, be sure to assign a unique name to each vial. Duplicate entries of the same reagent ID are not allowed in the inlet carousel (a message appears to notify you that the reagent is already in the carousel). Multiple phosphate buffer vials in the inlet carousel can be given names as such as PO4_RUN for run buffer, and PO4_PREP for prep reagent, PO4_DIP for dip reagent.

Define Configuration PEPTIDES2 -- Screen 2

Define Inlet Carousel

Reagent: NITROGEN gas purge

Sample: bradykinin + a-MSH

Empty

Pos No.	Type	ID/Description
1	Reagent	P04 PREP 148-5010 pho
2	Reagent	P04 RUN 148-5010 pho
3	Sample	Peptide Calibrator
4		
5		
6		
7		
8		

Reagent ID	Description	pH
IEFMobilizer 148-5030	IEF mobi	10.5
NITROGEN	gas purge	
PCR+U PREP	148-5025 PCR buff	8.3
PCR+U RUN	148-5025 PCR buff	8.3

Inlet Carousel Positions

Continue Cancel Clear

Reagents

Figure 5-23 Define Configuration Screen 2—Inlet Carousel

When you have completed vial assignments for the inlet carousel, click **Continue** to move to Define Configuration-Screen 3. Clicking **Cancel** returns you to the main BioFocus screen.

Outlet Carousel (Screen 3)

Figure 5-24 shows a Define Configuration—Screen 3 example. This screen allows you to designate what each vial contains and its position in the outlet carousel.

Define Configuration PEPTIDES2 -- Screen 3

Define Outlet Carousel

Reagent

Fraction

Waste

Empty

Pos No.	Type	ID/Description
1	Waste	
2	Reagent	P04 RUN 148-5010 pho
3		
4		
5		
6		
7		
8		

Reagent ID	Description	pH
ACID WASH	148-5036 acidic w	1.6
BASE WASH	148-5035 basic wa	12.0
BORATE PREP	148-5023 protein	8.5
BORATE RUN	148-5023 protein	8.5

Outlet Carousel Positions

OK Cancel Clear

Reagents

Figure 5-24 Define Configuration Screen 3—Outlet Carousel

→Tip:

Duplicate entries of the same Reagent ID are not allowed in the outlet carousel (a message appears to notify you that the reagent is already in the carousel). If you wish to use multiple vials for the same reagent in the outlet carousel, be sure to assign a unique name to each vial. For example, multiple phosphate buffer vials in the outlet carousel might be given a designation such as PO4_RUN, PO4_RUN_A, or PO4_RUN_B. Because fractions will be linked to methods later in the program, you may have duplicate entries of any fraction ID placed anywhere in the outlet carousel.

Reagents, waste vials, and empty vials are assigned in the same manner described for the inlet carousel. Click the appropriate button, then double-click the desired outlet carousel position. At least one reagent vial and one waste vial must be assigned to positions in the outlet carousel. A fraction vial is assigned in the manner of an inlet carousel sample vial—simply type in the name for each fraction you plan to collect. Up to 30 fraction vials or up to 31 waste vials may be assigned. Unlike sample vials, fraction and waste vials do not have to be contiguous. If multiple fractions are being collected, the carousel will rotate so that fraction vials are used in the order of programmed collection times. If you wish to use more than one waste vial for your analyses, be sure to enter them in your configuration. Multiple waste vials can be assigned during programming of an Automation Sequence.

When you have completed vial assignments for the outlet carousel, click **OK**. The configuration you just defined will be saved and a View Configuration screen (see Figure 5-25) will appear for your review automatically.

The screenshot shows a software window titled "View Configuration CZE". It contains two main sections: "INLET CAROUSEL POSITIONS" and "OUTLET CAROUSEL POSITIONS", each with a table of assignments. Below these is a "CARTRIDGE DATA" section with various fields and a row of buttons at the bottom.

INLET CAROUSEL POSITIONS			OUTLET CAROUSEL POSITIONS		
Pos No.	Type	ID/Description	Pos No.	Type	ID/Description
1	Reagent	PO4_PREP 140-5010 pho	1	Waste	
2	Reagent	PO4_RUN 140-5010 pho	2	Reagent	PO4_RUN 140-5010 pho
3	Sample	Peptide Calibrator	3		
4			4		
5			5		
6			6		

CARTRIDGE DATA					
Catalog Number	140-3831	Serial Number	PEPTID		
Length	24 cm	Diameter	25 mm	Coated	Y
Use Count	0	Active [Y/N]	Y		

Buttons: Select, Define, Edit, Browse, Exit

Figure 5-25 View Configuration Screen

Editing Configurations

You can make changes by clicking the **Edit** button. This will take you back through the series of screens just described. To change the configuration name, you must click **Define**. Clicking **Browse** returns you to a list of all configurations currently programmed. See Figure 5-26.

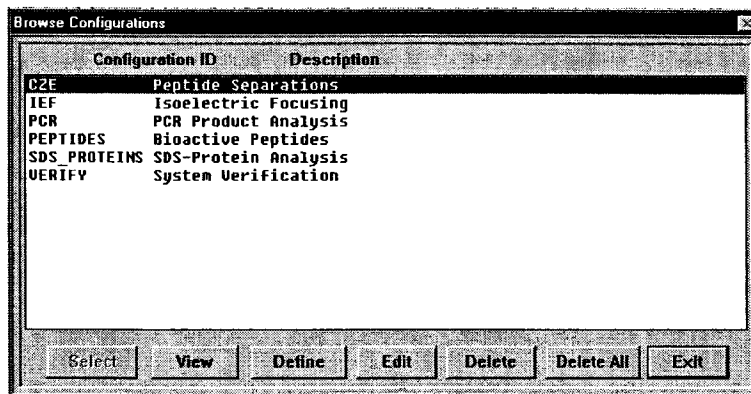


Figure 5-26 Browse Configurations Screen



Print Requests Button

Printing Configurations

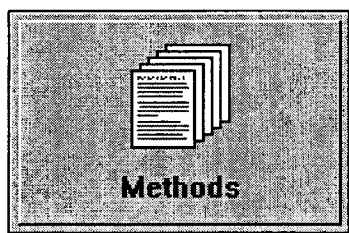
If you wish to print a hard copy of your configuration, you can do so from the **Print Requests** button on the toolbar (this button becomes inactive during the run).

➔ **Tip:** At this point, you may wish to load your carousels with run buffer, sample, internal standard, prep reagent, waste, and fraction vials, using the screen (or the hard copy of your configuration) as a reference. When completed, clicking **Exit** returns you to the main screen.

Defining Methods

A Method defines the operating parameters (run conditions) of the instrument.

The information in the following paragraphs describes defining absorbance methods. Refer to “LIF Detector Setup” on page 52 for LIF method definition procedures. The basic difference between absorbance and LIF methods is in the way you set up the detectors. Both methods use the same procedures for such things as choosing reagents, sample injection, setting up gradient methods, and defining automation sequences.



Methods Button

NOTE: The Detector column in the Browse Methods screen shows whether the method is absorbance (A) or fluorescence (F).

NOTE: If you are currently in absorbance mode, you cannot edit an LIF method that is linked to an LIF automation sequence.

Click Define to move to the Define Method screen.

Click Edit to make changes to a previously-defined method.

Defining Absorbance Methods

Clicking the **Methods** button on the main screen produces the Browse Methods screen (you can also access this screen from the menu bar). See Figure 5-27.

Bio-Rad supplies reference method templates with the instrument. The CZE, IEF and SHUTDOWN templates are noneditable, nondeletable and nonautodeletable from the database.

Method ID	Type	Detector	Method Description
CZE_TEMPLATE	CZE	A	peptide method
CZE_TMPL_LIF	CZE	F	peptide method
EQUILIBRATE	CZE	A	Equilibration Method
FINAL	CZE	A	Terminates Autosequence
IEF_TEMPLATE	IEF	A	IEF method
IEF_TMPL_LIF	IEF	F	IEF method
PCR	CZE	A	PCR Product method
SDS_PROTEIN	CZE	A	SDS Protein method
SHUTDOWN	CZE	A	terminates autosequence
SHUTDOWN_LIF	CZE	F	terminates autosequence
VERIFY	CZE	A	Verification Method

Buttons: Define, Edit, Delete, Delete All, Exit

Annotations:

- List of methods in database.
- Click Delete to delete a method that you highlighted.
- Click Delete All to delete all methods listed.
- Click Exit to return to the main screen.

Figure 5-27 The Browse Methods Screen

The PCR and SDS_PROTEIN method templates are also noneditable and nonautodeletable, but are singly-deletable unless they are linked to other record(s) in the database. You may also use any user-defined method in the database as a template.

Defining a New Method

To define a new method, first click to select the appropriate template then click the **Define** button. In this example, the CZE Template has been selected. Clicking the **Define** button will bring up the Define Method screen with the CZE template parameters in place. Refer to Figure 5-28.

Method ID and Description entries.

Run Parameters selection

Preparation Cycle Parameters

Injection settings

Detector settings

Timed Events settings

Cycle	Pre/Post	Inlet	Outlet	Pressure	Duration	Units
1	Pre-inj	INTERNALSTD	CZERUN	Low	10	psi*sec

Figure 5-28 Define Method Screen

→ Tip:

Careful selection of the method used as a template will minimize the amount of data entry necessary to define a new method.

Key to the Define Method Screen

Define Method

Method ID: CZE_TEMP2

Description: peptide method

Run Parameters:

◀ Voltage Mode Run Voltage: 10.0

> Current Mode Current Limit: 58.0

Preparation Cycle Parameters:

Cycle	Pre/Post	Inlet	Outlet
1	Pre-Inj	P04_PREP	Waste

Add Insert

Injection: Pressure, 20 (psi * sec)

Detector: Single Wavelength Mode

Timed Events

OK Inlet Reagent: P04_RUN

Cancel Outlet Reagent: P04_RUN

kV Polarity: ◀ + → - ▶ + + +

µA Capillary Temperature: 20 °C

Run Time: 15:00 minute[s]

1 cycle(s) defined

Inlet	Pressure	Duration	Units
Waste	High	60	sec

Delete Delete All

Enter unique method ID; up to 12 characters

Enter method description if desired

Select voltage or current mode

Set run voltage and current values

Double-click to bring up Pre/Post dialog box

Double-click to bring up inlet reagent box

Click to insert a cycle

Click to add a cycle

Click to select inlet vial reagent

Click to select outlet vial reagent

Toggle to select polarity

Enter capillary temperature

Enter run time

Enter duration for purge cycle

Double-click to bring up pressure

Figure 5-29 Define Method Screen Detail

Entering Method ID and Description

Each new method must be given a unique ID of up to 12 characters in length with no internal blank spaces. Enter a description for the new method in the Method ID box provided.

Selecting Inlet and Outlet Reagents

If you click the **Inlet Reagent** or **Outlet Reagent** button, a list box appears showing all of the reagents currently in the reagent database library. Any reagent can be selected by double-clicking it or by clicking it and then clicking the **OK** button. See Figure 5-30.

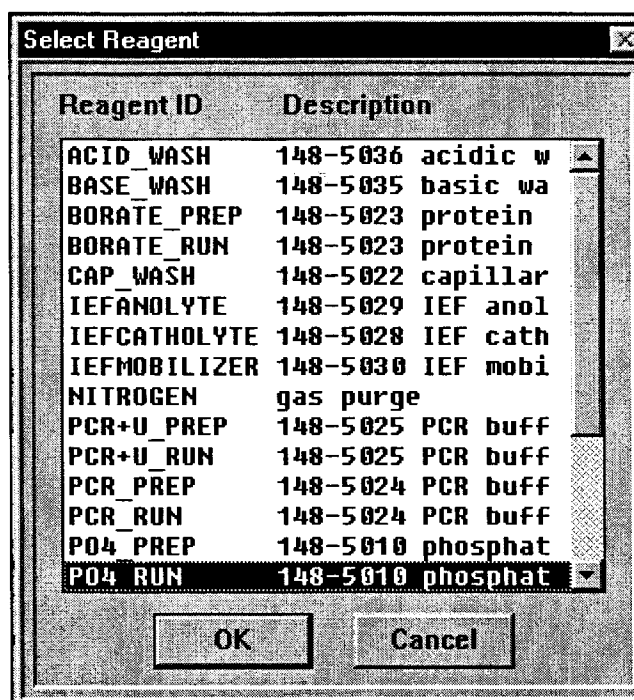


Figure 5-30 Select Reagent Screen

→Tip:

If you wish to run the same samples with different run buffers, you must create a new method for each type of run reagent. These can later be programmed as different run groups within an automation sequence.

After you have selected both reagents, their Reagent ID's should appear next to Inlet and Outlet Reagent buttons on the Define Method screen. Be sure that the inlet and outlet run buffers are compatible.

NOTE: Polarity direction is from the inlet (high voltage) to the outlet (ground) electrode. While a Run Voltage value up to 30 kV is permitted, the user should always take into consideration the capillary size and ionic strength of the buffer before entering a value. Excessively high field strengths may result in damage to the capillary.

Entering Run Parameters

Use the following procedures to set Run Parameters:

Selecting Voltage or Current Mode

Entering Run Voltage and Limit Values

If voltage mode is selected, you will enter run voltage and current limit. If current mode is selected, you will enter run current and voltage limit.

Selecting Polarity

Select polarity by clicking the + → - or - → + button.

Setting Capillary Temperature

Enter capillary temperature, from 15 through 65° C.

Entering Run Time

Enter run time in box provided. Note that the software will not accept run times less than 1 minute or greater than 120 minutes.

Entering Preparation Cycle Parameters

The Preparation Cycle Parameters table defines the number of steps for capillary washes, run buffer filling of the capillary, rinsing of the electrode and outer surface of the capillary (dip cycle) before or after sample injection, and the conditions for injecting internal standards without manual spiking of the sample (Multi Inject).

Parameters that define a prep cycle include type of cycle (Pre- or Post-inject), inlet and outlet reagents (or waste), pressure (high or low), and cycle duration.

Setting Cycle

1. Click the box under Cycle to add the first preparation cycle. This brings up default parameters for run cycle 1.
2. Change the parameters to match your method by following the steps outlined below.

NOTE: All Pre Inject cycles must be scheduled before all Post Inject cycles.

Setting Pre/Post Injection Mode

The prep cycles can either be scheduled before (Pre-injection) or after (Post-injection) sample injection. If you want to change the cycle type for the cycle being defined, double-click Pre-inj. This produces the selection box shown in Figure 5-31. Make your selection by clicking your choice. Your selection will now automatically appear in the Pre/Post Column.

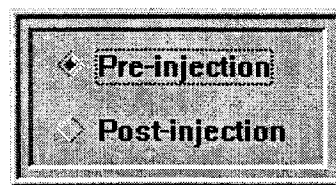


Figure 5-31 Pre-Injection Selection Box

1. Click Injection to set injection mode. This brings up a selection box.
2. Click Pre-injection or Post-injection as required. Note that the software will not allow you to assign a post-injection cycle prior to a pre-injection cycle.

You may not change a Pre Inject prep cycle to Post Inject if at least one Pre Inject cycle follows it. To make such a change, start with the final Pre Inject cycle and reassign it as a Post Inject cycle. Then, move to the cycle that you wish to be Post Inject, reassigning each one in turn as a Post Inject cycle.

You may not change a Post Inject prep cycle to Pre Inject if at least one Post Inject cycle precedes it. To make such a change, start with the first Post Inject cycle and redesignate it as a Pre Inject cycle. Then, move to the cycle that you wish to be Pre Inject, reassigning each one in turn as a Pre Inject cycle.

NOTE: Bio-Rad 150 μm OD capillaries should only be used with a Run Voltage under 20 kV. For operation above 20 kV, Fluorinert should be used as the capillary coolant.

Setting Inlet and Outlet Purge Reagents

1. Double-click the box under Inlet. This brings up a reagent list screen.
2. Choose the inlet reagent and double-click the name.
3. Double-click the box under Outlet. This brings up a reagent list screen. See Figure 5-34.
4. Choose the outlet reagent and double-click the name.

Setting Pressure

You may designate each cycle to be either a purge-like High Pressure cycle or an injection-like Low Pressure cycle. High pressure cycles (at 100 psi) are in seconds whereas low pressure cycles (at 5 psi) are in $\text{psi} \cdot \text{sec}$. If you want to change the pressure type for the cycle being defined, double-click it. This opens the selection box shown in Figure 5-32.

1. Double-click the box under Pressure. This brings up the selection box shown in Figure 5-32.
2. Select High or Low as required.

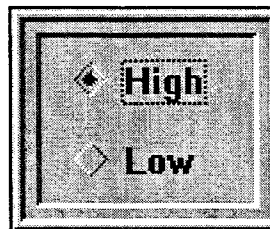


Figure 5-32 Pressure Injection Selection Box

NOTE: The units for the pressure cycle will automatically switch between sec (for high pressure) and $\text{psi} \cdot \text{sec}$ (for low pressure) as you make your selection.

Make your selection by clicking your choice. Your selection will now appear in the Pressure column.

Pre Inject cycles at High Pressure are recommended for capillary washes, filling of the capillary with run buffer, filling of the entire capillary with sample during ITP separations, or rinsing the electrode and the outer surface of the capillary (dip cycle) *before* sample injection.

Setting Duration and Units

If you wish to select a prep duration different than the default, click the box under Duration and enter a duration for this cycle by typing in the number.

1. Enter a duration in the box under Duration. Time units automatically match your pressure setting, either seconds for high pressure or $\text{psi} \cdot \text{sec}$ for low pressure.

Dip Cycles

If you wish to include dip cycles (of zero duration) after each injection step to avoid carryover, you should use run buffer both as the Inlet and the Outlet purge buffers and set a duration of zero seconds. Make sure to include multiple, uniquely-labeled run buffer vials in the reagent data-base and your configuration. Refer to Figure 5-33.

The screenshot shows the 'Define Method' window with the following settings:

- Method ID:** GZE_TEMPLATE
- Description:** peptide method
- Run Parameters:**
 - Voltage Mode: Run Voltage 10.0 kV
 - Current Mode: Current Limit 50.0 μ A
 - Polarity: $\leftarrow + + -$
 - Capillary Temperature: 20 $^{\circ}$ C
 - Run Time: 15.00 minute[s]
- Preparation Cycle Parameters:** 3 cycle[s] defined

Cycle	Pre/Post	Inlet	Outlet	Pressure	Duration	Units
1	Pre-inj	P04_PREP	Waste	High	60	sec
2	Pre-inj	IEFANOLYTE	Waste	Low	5	psi*sec
3	Pre-inj	SDS DIP1	Waste	Low	0	psi*sec
- Buttons:** Add, Insert, Delete, Delete All
- Injection:** Pressure: 20 {psi * sec}
- Detector:** Single Wavelength Mode, 488 nm
- Timed Events:**

Figure 5-33 Define Method Screen, Dip Cycle Added

Preparation Cycle Tips

We recommend that the current limit (constant voltage mode) or voltage limit (constant current mode) be set at values approximately 50% higher than the actual expected run current or voltage. However, to preserve a stable baseline in constant voltage mode, the current limit should be set at 10 μ A or greater, even when actual run current is much lower.

Cartridge temperature range is 15 $^{\circ}$ to 65 $^{\circ}$ C. Enter a duration (minutes or seconds) in the box under Duration. Time units automatically match your pressure setting, either seconds for high pressure or psi*sec for low pressure.

Cartridge temperature range is 15 $^{\circ}$ to 40 $^{\circ}$ C. Prep cycle information is entered in the Preparation Cycle Parameters table. Other parameters are defined on other screens by clicking the **Inlet Reagent**, **Outlet Reagent**, **Injection** and **Detector** buttons. These areas can be accessed in any order. They can also be altered at any time before the user clicks the Define Method screen **OK** button, indicating that all parameters have been defined and the complete method should be saved. Prep cycle information is entered in the Preparation Cycle Parameters table. Other parameters are defined on other screens by clicking the **Inlet Reagent**, **Outlet Reagent**, **Injection** and **Detector** buttons. These areas can be accessed

in any order. They can also be altered at any time before the user clicks the Define Method screen **OK** button, indicating that all parameters have been defined and the complete method should be saved.

NOTE: All Pre Inject cycles must be scheduled before all Post Inject cycles.

You may not change a Pre Inject prep cycle to Post Inject if at least one Pre Inject cycle follows it. To make such a change, start with the final Pre Inject cycle and reassign it as a Post Inject cycle. Then, move to the cycle that you wish to be Post Inject, reassigning each one in turn as a Post Inject cycle.

You may not change a Post Inject prep cycle to Pre Inject if at least one Post Inject cycle precedes it. To make such a change, start with the first Post Inject cycle and reassign it as a Pre Inject cycle. Then, move to the cycle that you wish to be Pre Inject, reassigning each one in turn as a Pre Inject cycle.

Adding a Prep Cycle

You may add a prep cycle at the end of existing cycles or at the beginning if you have not define a cycle previously. Click the **Add** button; a new cycle appears for definition.

Inserting a Prep Cycle

If you have two or more prep cycles defined, you may insert a cycle between existing cycles. Use the mouse to highlight the cycle *after* the cycle you wish to insert. Click the **Insert** button; a new cycle appears above the cycle you highlighted.

Deleting a Prep Cycle

To delete a prep cycle, highlight it and click the **Delete** button. A dialog box appears to allow you to confirm the deletion or cancel it.

Deleting All Prep Cycles

Alternatively, for either situation, you may delete all prep cycles by clicking **Delete All** and starting over.

Changing a Prep Cycle

If you want to change the Inlet prep reagent for the cycle being defined, double-click it. This produces a Select Reagent list box, displaying all reagents in the reagent database. Double-click your choice, and the new inlet reagent now appears under Inlet.

The default position for the Outlet prep reagent is Waste. If you wish to change this entry to another reagent, follow the steps above. After assigning the new outlet reagent, if you change your mind and want to go

back to Waste, you can do so by double-clicking this cell. This will produce a message asking if you wish to designate a Waste as the outlet prep reagent for this cycle. Click **Yes** to enter Waste as the outlet reagent. If you click **No**, the Select Reagent list box will reappear for you to make your selection. See Figure 5-34. Either select a new reagent or click **Cancel** to return to the Define Method screen.

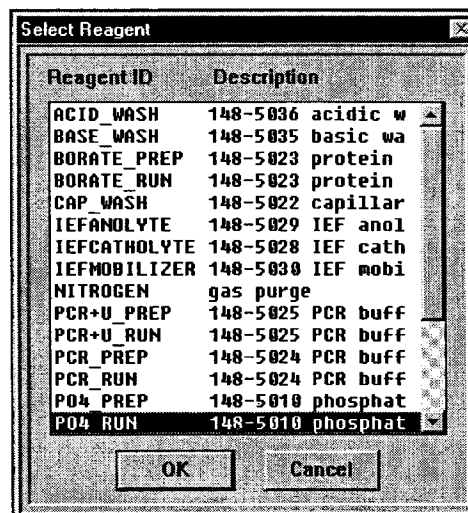


Figure 5-34 Select Reagent Screen

Maximum Number of Prep Cycles

Within a method, you can define up to a total of 12 prep cycles. To define a second prep cycle, click **Add** (or on scroll bar). This will add a second row to the Preparation Cycle Parameters table for you to define the new cycle. You cannot add a third prep cycle to this table until the second cycle's information is complete. You can either select the inlet reagent or delete the highlighted cycle by clicking **Delete**.

→ Tip: Pre-injection or Post-injection cycles at Low Pressure can be used for injecting internal standards without manual spiking of the sample to avoid sample contamination or sample wastage (Multi-injection).

Post Inject cycles scheduled either at high or low pressure for 0 duration (dip cycle) can be used to rinse the electrode and the outer surface of the capillary *after* sample injection to eliminate sample carryover into run buffer vials. Similarly, Pre Inject cycles scheduled at either high or low pressure for 0 duration can be used to rinse electrode and capillary *before* sample injection to eliminate purge solution carryover into sample vials.

You can also insert a prep cycle before the highlighted cycle by clicking **Insert**.

Clicking **Delete All** deletes all prep cycles and clears the Preparation Cycle Parameters table. You can then add a prep cycle by clicking **Add** (or **Insert**). The new prep cycle will come up with the inlet reagent undefined. You must select an inlet reagent to continue.

Multi Inject

The Multi Inject feature of the software allows you to inject internal standards without manual spiking of the sample, avoiding contamination and/or wastage of precious samples. If you wish to use Multi Inject, you must first enter the internal standard(s) you plan to inject in the reagent database library. Then, assign the internal standard(s) vial(s) to position(s) in the inlet carousel of the desired Configuration before you create a Method.

In CZE, best separation efficiency is achieved when the internal standard is prepared in the run buffer and injected using a Pre Inject cycle at Low Pressure (5–10 psi*sec). Sample should be prepared in 1:10 (v/v) diluted run buffer. For the Multi Inject cycle, enter the Internal Standard as the Inlet and the run buffer as the Outlet reagent in the Preparation Cycle Parameters table.

Sample Injection Selection

By clicking **Injection**, the type of sample injection and associated parameters can be defined from this screen. There are three options: Electrophoretic, Pressure, and No Injection. Refer to Figure 5-35.

→ Tip:

Electrophoretic injection may provide a more narrow starting zone than pressure injection when working with wide diameter capillaries.

Electrophoretic

If you select **Electrophoretic** injection, you must type in or click the values for mode, voltage, current, and inject time. The current or voltage limit should be set at a value about 50% greater than the expected run value. For best accuracy, current limits should be set at 50 μ A or higher in constant voltage mode.

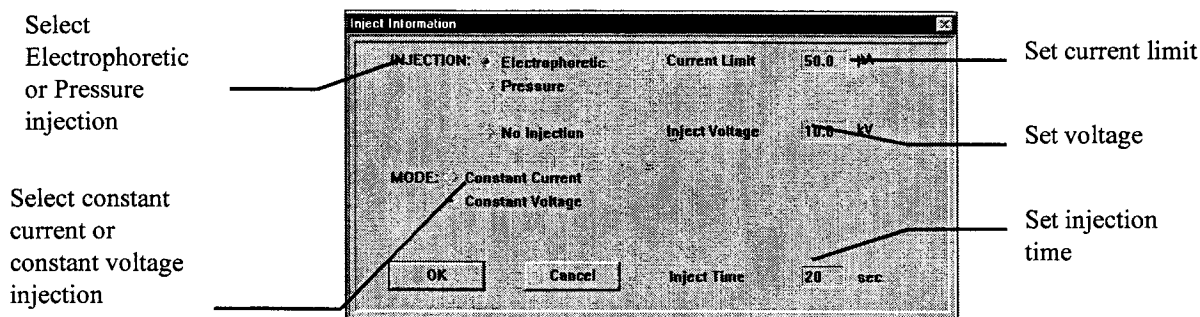


Figure 5-35 Inject Information Screen—Electrophoretic Injection

Pressure

If you select **Pressure** injection, you must enter a Pressure*Time Injection Constant in a box at the lower right corner of the screen (see Figure 5-36).

To compensate for small fluctuations in injection pressure during a short load time, you simply program a single pressure * time (psi*sec) constant. If you wish to inject a sample at 5 psi for 4 seconds, you enter a value of 20.

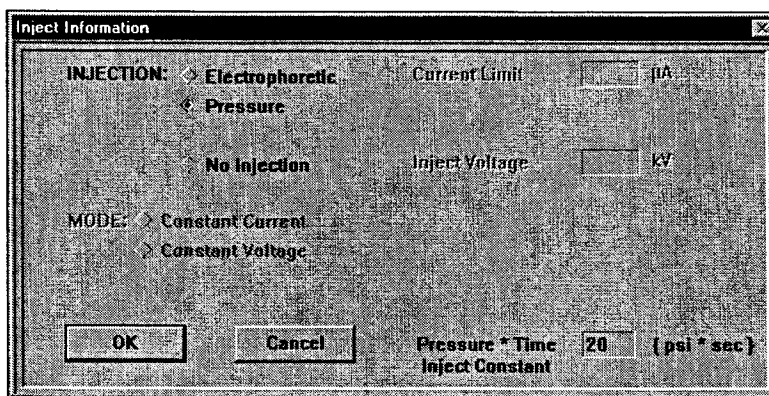


Figure 5-36 Inject Information Screen—Pressure Injection

No Injection

You may select **No injection** for the following applications: to monitor the baseline under high voltage conditions for a noise signature, to measure the rate of electroendosmosis, or to schedule a shutdown method at the end of your automation sequence.

When you complete setting inject parameters, click **OK** to return to the Define Method screen. The new injection parameters now appear next to the Injection button.